together further downstream at the location of the detector. Treating the light as a wave, as in the description by Maxwell, we expect—and, indeed, we find—an interference pattern on the screen. The length of the journey to all but the center point on the screen is slightly different for the left and right routes and so while the left beam might be reaching a peak at a given point on the detector screen, the right beam might be reaching a peak, a trough, or something in between. The detector records the combined height of the two waves and hence has the characteristic interference pattern.

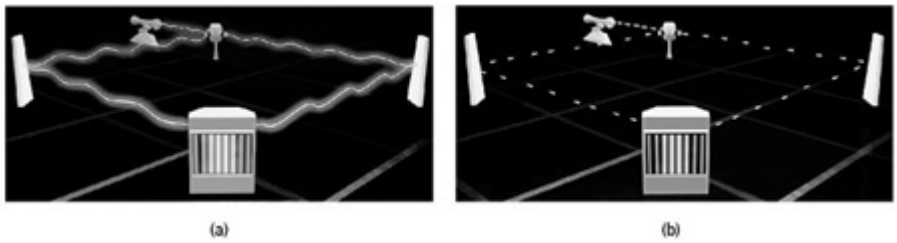


Figure 7.1 (a) In a beam-splitter experiment, laser light is split into two beams that travel two separate paths to the detector screen. (b) The laser can be turned down so that it fires individual photons; over time, the photon impact locations build up an interference pattern.

The classical/quantum distinction becomes apparent as we drastically lower the intensity of the laser so that it emits photons singly, say, one every few seconds. When a single photon hits the beam splitter, classical intuition says that it will either pass through or will be reflected. Classical reasoning doesn't even allow a hint of any kind of interference, since there is nothing to interfere: all we have are single, individual, particulate photons passing from source to detector, one by one, some going left, some going right. But when the experiment is done, the individual photons recorded over time, much as in Figure 4.4, *do* yield an interference pattern, as in Figure 7.1b. According to quantum physics, the reason is that each detected photon *could* have gotten to the detector by the left route or by going via the right route. Thus, we are obliged to combine these two possible histories in determining the probability that a photon will hit the screen at one particular point or another. When the left and right probability waves for each individual photon are merged in this way, they yield the undulating probability pattern of wave interference. And so, unlike Dorothy, who is perplexed when the Scarecrow points both left and right in giving her directions to Oz, the